

Cry for AI: A Moss-Controlled AI Wearable for Resource-Aware Interaction

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Figure 1: Overview of Cry for AI. (Center) A user wearing the Cry for AI device and hardware components (Low-Right) screenshots of the web dashboard showing real-time AI resource-usage visualization and moss-condition tracking.

Abstract

Seemingly infinite AI requests rely on finite material resources like cooling water, computational electricity, and human labor, that largely remain invisible to end users. We present *Cry for AI*, a nature-controlled AI device that makes these dependencies tangible by giving living moss an access-control role over AI interaction. While prior work on sustainable AI has focused primarily on system-level efficiency or societal framing, few approaches address sustainability as an embodied, moment-to-moment interaction. *Cry for AI*

introduces a personalized AI resource-budgeting interface that situates sustainability within a human–AI–nature feedback loop. Users select voice-controlled modes with predefined resource budgets, and moss viability determines whether AI responses are enabled, limited, or refused. When dehydration crosses a threshold, access is restricted until a tear-activated sensor restores hydration. *Cry for AI* reframes AI not as an immaterial service but as an interaction contingent on finite resources, allowing living system to regulate and reshape everyday use.

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CCS Concepts

• Human-centered computing → Interaction devices; • Hardware → Impact on the environment.

Keywords

sustainable AI, wearable device, biomaterial

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1 Introduction

Generative AI systems are widely used for searching, drafting, coding, and decision support. However, their operation depends on material infrastructures that are rarely present at the interaction surface. Each prompt draws on electricity and water intensive data center cooling, as well as land and labor distributed across global supply chains [2, 9, 13]. Because these infrastructures are spatially and institutionally distant, the cloud-based AI services are experienced as immaterial, although it is grounded in energy systems, water systems, land, and labor. Recent work quantifies these dependencies, yet they rarely surface in everyday use [3, 11].

Within HCI, critical design and critical making often use artifacts to reveal what is usually hidden in socio-technical systems. Sustainability oriented work builds on this approach by foregrounding material lifecycles and how repeated micro decisions aggregate at scale. More-than-human perspectives further extend this stance by treating nonhuman entities not as passive resources but as participants with vulnerabilities and constraints that can shape interaction.

Motivated by these lines of inquiry, we designed *Cry for AI*, a nature controlled AI device that makes AI's resource dependencies experiential by placing a living moss interface in the interaction loop. *Cry for AI* integrates a conversational AI endpoint with a moss wearable, a shared water reservoir and humidification module, and a tear sensing element. Users interact with voice controlled AI modes that correspond to different resource budgets. Each prompt debits water from the reservoir to sustain moss viability through humidification. When the moss dries beyond a threshold, the system restricts access and can refuse to respond. Access is restored only when tears activate the sensor and resume water delivery, making ecological viability an explicit condition of AI availability.

Cry for AI treats user control as negotiation with constraints. It introduces deliberate friction by allowing refusal when ecological limits are reached, and it links prompting to cumulative resource demand through the ongoing maintenance of a living interface. Tears function as a circuit input that returns the body to the loop as both the recipient of AI support and a material contributor within a human–AI–nature resource exchange.

In this work, we contribute:

- (1) A bio-responsive AI interface that uses voice interaction and explicit resource constraints to mediate AI access.
- (2) An ecological access control framework in which a nonhuman living system can budget, limit, and refuse interaction within Human–AI loop.
- (3) A biodegradable wearable fabrication method that combines a PHA structure designed to sustain moss viability.

2 Related Works

2.1 Sustainable HCI

Sustainable HCI research has developed interactive approaches that connect everyday use to environmental impact. Central to this work is eco-feedback, which ties everyday interactions to energy use and material impact, and offers design methods for assessing digital services across lifecycles [3, 6, 11, 12]. Yet these approaches are largely informational, offering awareness rather than interaction mechanism that can directly shape action at the moment of use. Recent CHI researches quantify emissions from generative AI use in HCI workflows and provide tools for estimating and offsetting these footprints [1, 6, 7]. However, it remains largely carbon centered and researcher facing, with limited support for framing everyday natural resources in ways that resonate with end users and shape prompting decisions. Related research also examined the assumptions embedded in machine learning eco feedback metrics and evaluated user acceptance of carbon aware tradeoffs such as slower response times [5, 8]. Still, sustainability is framed as a configurable service attribute, leaving opportunities to develop end user feedback mechanisms that are more perceptible and that more strongly support reflection.

2.2 Nature as Interface

Recent work in HCI and design has begun to treat living and transforming materials as integral components of interactive systems, rather than as mere context or decoration. Giaccardi and Karana et al. [4] argue that the experiential qualities of materials themselves, such as softness, moisture, and change over time, should be treated as first-class interaction resources in HCI, while Tsaknaki and Fernaeus et al. [14] draw on wabi-sabi to value patina, decay, and natural transformation as aesthetic resources in interaction design. In parallel, plant-based interfaces demonstrate how living plants can serve as sensors or interactive surfaces: in *Botanicus Interacticus*, capacitive sensing turns plants into rich, touch-sensitive controllers [10], and subsequent work presented at CHI has explored how organic change and material aging can function as slow, ambient forms of feedback rather than discrete events [14]. Together, these strands suggest that growth, material change, and living agency can be treated as first-class interaction qualities. Our work builds on this trajectory by designing not only with plants themselves, but also with a bio-based hydrogel substrate that is intentionally grown together with moss, positioning the evolving living–material assemblage as the primary interface form.

3 Implementation

Cry for AI is a wearable system designed as a self-contained ecological interface in which living moss actively mediates access to AI agent functionality. To achieve this, we articulate two design goals (DG):

- **[DG 1]** Design a biodegradable on-body environment that supports moss vitality and enables its expressive agency.
- **[DG 2]** Develop an integrated sensing–actuation loop connecting moss conditions to the wearable circuit and cloud-based AI system.

To achieve these goals, the hardware integrates biodegradable materials with embedded electronics that sense moss conditions, regulate hydration, and gate AI responses based on environmental state (Figure 2). By embedding sensing, actuation, and computation on-body, the system makes ecological constraints directly operative within everyday AI interaction.

3.1 Moss Material Experiment

Given the central role of nature within our DG1, we conducted material research to explore how moss(a resilient, adaptable, and ecologically significant organism) can be integrated into a wearable interface. Our goal was to identify a natural material capable of expressing agency, responding to environmental change, and growing safely on or near the skin. Moss was selected for its symbolic and biological relevance: as one of the earliest plant forms, moss plays a key role in redistributing water in Earth ecosystems. Materially, its lack of roots, seeds, or flowers and its membrane-based water absorption make it well-suited for hydrogel-based environments.

For initial evaluation, we selected five moss types with diverse ecological properties:

- **Cushion Moss:** Forms dense moisture-retaining tufts on rocks and bark.
- **Marchantia (Umbrella Moss):** A fast-growing liverwort common in wet, shaded environments.
- **Peat Moss:** Highly absorbent sphagnum species with strong water-retention capacity.
- **Desert Moss:** Adapted to arid climates, capable of surviving prolonged dryness.
- **Süßwasserkang:** A freshwater fern used in aquaria, growing on submerged surfaces.

Using these species, we explored the viability of moss growth across multiple hydrogel matrix formulations (Figure 3). Microscope observations of dehydration and rehydration cycles provided insight into each species' tolerance and recovery patterns. We conducted two form-finding experiments: (1) extruding the hydrogel as yarn and planting moss along perforated channels, and (2) casting hydrogel into 3D-printed organic molds to create patterned growth surfaces.

Through these experiments, we identified PHA-printed surfaces combined with our final hydrogel formulation as the most supportive environment for moss vitality and integration into the wearable system.

Final Hydrogel Formulation Sodium Alginate 2% (w/v), Glycerin 12% (w/v), Gelatin 10% (w/v), Cushion Moss 5% (w/v), CaCl_2 5% (w/v).

For our prototype, we prepare 300 grams of biocomposite mix paste, which includes 6 grams of Sodium Alginate, 36 grams of Glycerin, 30 grams of Gelatin, and 15 grams of Cushion Moss. This mixture is submerged in a CaCl_2 solution for crosslinking, resulting in a sealed exterior that maintains a moist and breathable interior, thereby allowing the moss to thrive.

Our prototype employs a hand-extruded paste applied to a surface, which is subsequently cast and integrated with a 3D-printed skeleton. Rather than conventional materials such as PLA, we selected PHA, a biodegradable and bio-based polyester synthesized by microorganisms. The elastic and flexible properties of PHA allow it

to bend and conform to the contours of the human body, providing an eco-friendly alternative to traditional petroleum-based plastics and ensuring that the entire wearable remains biodegradable.

3.2 Electronics

The electronic system consists of three core components: (1) embedded AI, (2) moss condition sensing system, and (3) water circulation and hydration mechanism.

The center of the system is an XIAO ESP32-S3 microcontroller, responsible for sensor integration, power management, and communication with AI server through wifi. The microcontroller interfaces with an on-device microphone for user input and a speaker for AI-generated audio responses, enabling fully hands-free interaction.

The condition of the moss is monitored using four distributed humidity sensors(Adafruit HTU21D-F) positioned on the wearable surface to capture localized moisture levels. When the measured humidity falls below a predefined threshold (H_{min}), the system disables AI responses, causing the device to explicitly refuse user queries until environmental conditions recover. This mechanism ensures that AI availability remains contingent on the vitality of the living substrate.

Hydration is facilitated through a tear-responsive watering system. Conductive textile sensors placed beneath the user's eye detect changes in conductivity associated with tear fluid. When conductivity drops below a calibrated threshold (C_{tear}), a micro water pump(Micro Peristaltic Pump) embedded in the back of the necklace activates, circulating water through internal tubes. The water is absorbed by a cotton stick and connected to the ultrasonic humidifier, which supplies moisture to the moss and restores system operability.

3.3 Software

Cry for AI runs a hybrid software architecture composed of embedded firmware on the XIAO ESP32-S3¹ and a web-based visualization interface. The embedded software manages sensing, power control, threshold-based decision logic, and communication with cloud-based AI services. A web dashboard visualizes real-time moss humidity levels and AI usage data log.

AI responses are generated via the Gemini API², with speech output synthesized using ElevenLabs³. To foreground resource awareness, the system provides three distinct interaction modes, each with different prompt structures and model configurations corresponding to varying computational and energy costs. Through explicit mode selection, the wearable has the potential to shift AI interaction toward a budgeting-oriented paradigm, instead of treating computational resources as implicitly unlimited.

Users invoke interaction modes through voice commands by the on-device microphone, explicitly specifying the desired resource level at the start of each request. This spoken mode selection makes computational cost legible at the moment of interaction and situates AI usage as a deliberate, contextual choice rather than a default action(see Table 1). By embedding resource negotiation directly

¹https://wiki.seedstudio.com/xiao_esp32s3_getting_started/

²We use Google's Gemini API for cloud-based language model inference with configurable model parameters. <https://ai.google.dev/gemini-api/docs>

³ElevenLabs provides text-to-speech generation compatible with the device's audio pipeline. <https://elevenlabs.io/>

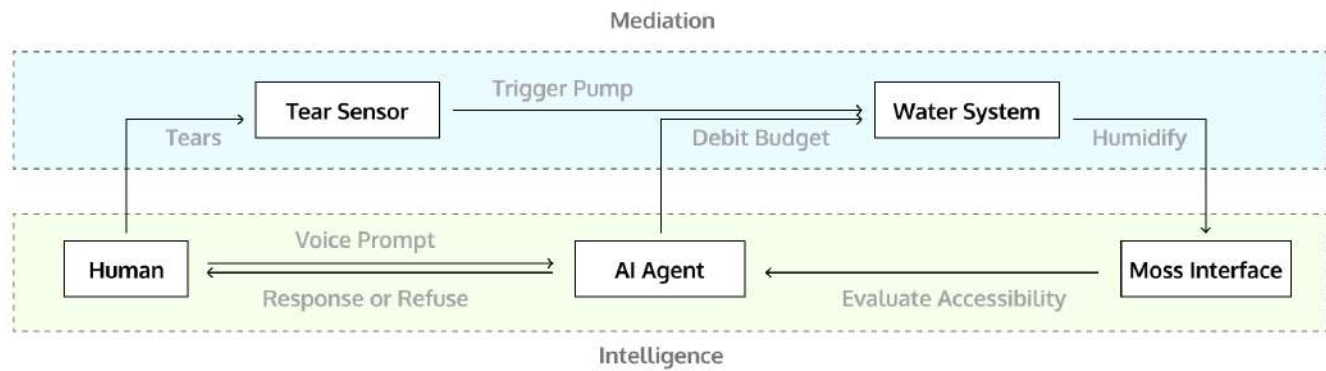


Figure 2: Interaction loop illustrating how the hardware (moss sensing, hydration, and on-device computation) and software (mode selection, AI inference, and resource tracking) operate together to regulate AI availability based on ecological conditions.

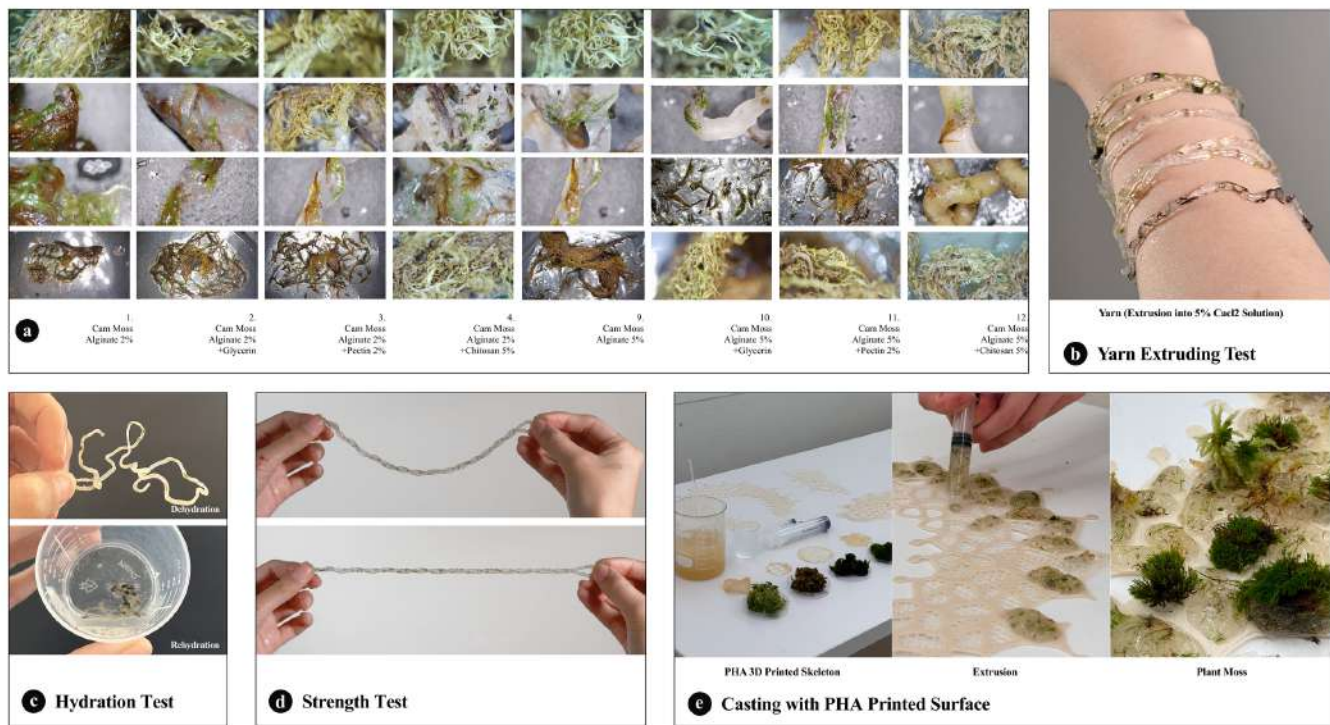


Figure 3: Overview of the material research with moss using hydrogel substrates and controlled hydration experiments. (a) Recipe and fabrication process for hydrogel yarn extrusion, moss types used, and microscope images showing hydration and dehydration states across samples. (b) Yarn extrusion tests comparing four moss types, with close-up views of moss seeded onto extruded hydrogel. (c) Dehydration and re-hydration tests of extruded hydrogel yarn. (d) Tensile strength testing of the hydrogel yarn. (e) Casting experiments using 3D-printed organic patterns, showing hydrated states and separation of dehydrated cast surfaces from the mold.

into natural language input, the system integrates sustainability considerations into everyday conversational AI use.

The dashboard records daily AI interactions by mode and estimates water and electricity consumption by multiplying response

duration with each mode's resource coefficients, producing a summarized usage report (see Figure 1). This translates abstract infrastructural costs into a personalized and embodied sustainability narrative. Through this system, *Cry for AI* situates environmental dependency as an active participant in AI interaction rather than a hidden externality.

Table 1: Cry for AI Interaction Modes, Configurations, and Example Cases

Mode	Temp	Max Tokens	Reasoning Effort	Example Case
Green	0.3	50	Minimal	"Hey Moss, Green Mode. How do I make pasta?"
Agent	0.6	120	Medium	"Hey Moss, Agent Mode. Search my university emails and draft a reply."
Superpower	0.8	250	High	"Hey Moss, Superpower Mode. Research sustainable AI papers and write a report, adding it to my document."

4 Conclusion

Cry for AI contributes a moss-based wearable interface that positions a living material as an active regulator of AI access, reframing sustainability as an embodied, interaction-level concern. Looking forward, this work opens several design and research directions within sustainable HCI. In terms of material interaction, future iterations could incorporate color-changing or responsive substrates to make moss microstates more legible on the body. Beyond a personal device, the system also has potential as a social platform supporting community practices around AI resource budgeting, enabling users to share usage logs, compare strategies for maintaining moss health, and collectively reflect on sustainable AI habits. Beyond the wearable itself, this project highlights how living materials can express agency within computational systems, pointing toward broader possibilities for integrating nature's intelligence into sustainable AI design and everyday ecological action. Ultimately, *Cry for AI* suggests how rethinking AI through more-than-human relations can open new pathways for ecological accountability in interaction design.

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